

Assignment – 3

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Personal Expense Tracker

Objective

The objective of this project is to develop a Personal Expense Tracker using Python. The program helps users record their daily expenses, categorize them, and view spending summaries. It also stores expense data in a file so that the information is available even after the program is closed and reopened.

Tools and Technologies Used

- Python 3
- JSON File Handling
- Functions
- Lists and Dictionaries

Project Description

The Personal Expense Tracker is a menu-driven Python application that allows users to manage their expenses efficiently. Users can add new expenses by entering the amount, category, and date. The program stores all expense records in a JSON file and loads them automatically when the application starts.

The application provides various summaries such as:

- Viewing all expenses
- Category-wise spending summary
- Overall spending summary
- Monthly spending summary

This helps users analyze their spending habits and maintain financial records.

Features

1. Add Expense
2. View All Expenses
3. Category-wise Summary
4. Overall Spending Summary
5. Monthly Spending Summary

6. Save and Load Data Using File Handling

Data Structure Used

The project uses a list of dictionaries to store expense records.

Example:

```
{  
"amount": 500,  
"category": "Food",  
"date": "2026-06-13"  
}
```

File Handling

The project uses a JSON file named "expenses.json" for data persistence.

- `json.dump()` is used to save expense records to the file.
- `json.load()` is used to load previously saved records when the program starts.

This ensures that data is not lost after the program is closed.

Functions Used

1. `load_expenses()` – Loads expense data from the JSON file.
2. `save_expenses()` – Saves expense data to the JSON file.
3. `add_expense()` – Adds a new expense record.
4. `view_expenses()` – Displays all expenses.
5. `category_summary()` – Displays category-wise spending.
6. `overall_summary()` – Displays total spending.
7. `monthly_summary()` – Displays monthly expense summaries.
8. `main()` – Controls the menu-driven program.

Conclusion

The Personal Expense Tracker successfully records and manages user expenses. The project demonstrates the use of Python programming concepts such as functions, lists, dictionaries, loops, conditional statements, and file handling. The program provides a simple and effective way to track personal spending while ensuring data is stored permanently using JSON file handling.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell + - [ ] [ ] ... [ ]
PS C:\Users\denny\OneDrive\Desktop\ExpenseTracker> python main.py
1. Add Expense
2. View All Expenses
3. Category Summary
4. Overall Spending
5. Monthly Summary
6. Exit
Enter your choice: 1
Enter amount: 500
Enter category: food
Use today's date? (y/n): y
Expense added successfully!

===== PERSONAL EXPENSE TRACKER =====
1. Add Expense
2. View All Expenses
3. Category Summary
4. Overall Spending
5. Monthly Summary
6. Exit
Enter your choice: 1
Enter amount: 200
Enter category: transport
Use today's date? (y/n): y
Expense added successfully!

===== PERSONAL EXPENSE TRACKER =====
1. Add Expense
2. View All Expenses
3. Category Summary
4. Overall Spending
5. Monthly Summary
6. Exit
Enter your choice: 2

----- All Expenses -----
Amount: ₹500.0 | Category: food | Date: 2026-06-12
Amount: ₹200.0 | Category: transport | Date: 2026-06-12
```

```
PS C:\Users\denny\OneDrive\Desktop\ExpenseTracker> python main.py
===== PERSONAL EXPENSE TRACKER =====
1. Add Expense
2. View All Expenses
3. Category Summary
4. Overall Spending
5. Monthly Summary
6. Exit
Enter your choice: 4

Total Spending: ₹ 700.0

===== PERSONAL EXPENSE TRACKER =====
1. Add Expense
2. View All Expenses
3. Category Summary
4. Overall Spending
5. Monthly Summary
6. Exit
Enter your choice: 4

Total Spending: ₹ 700.0

===== PERSONAL EXPENSE TRACKER =====
1. Add Expense
2. View All Expenses
3. Category Summary
4. Overall Spending
5. Monthly Summary
6. Exit
Enter your choice: 5

----- Monthly Summary -----
2026-06: ₹700.0
```

The image shows a Visual Studio Code editor interface. The Explorer sidebar on the left shows a project named 'EXPEN...' with two files: 'expenses.json' and 'main.py'. The main editor area displays the 'expenses.json' file, which contains a JSON array of two objects. The first object has 'amount': 500.0, 'category': 'food', and 'date': '2026-06-12'. The second object has 'amount': 200.0, 'category': 'transport', and 'date': '2026-06-12'. Below the editor is a terminal window with a PowerShell prompt. The user has run 'python main.py' and entered '5' for the choice. The program has printed a 'Monthly Summary' for 2026-06 showing ₹700.0. It then displays a menu for the 'PERSONAL EXPENSE TRACKER' with options 1-6. The user entered '6' to exit, and the program printed 'Thank you for using Expense Tracker!'.

```
{} expenses.json X main.py
{} expenses.json > ...
1  [
2      {
3          "amount": 500.0,
4          "category": "food",
5          "date": "2026-06-12"
6      },
7      {
8          "amount": 200.0,
9          "category": "transport",
10         "date": "2026-06-12"
11     }
12 ]

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS powershell + - [ ] ... | [ ] [ ] X
PS C:\Users\denny\OneDrive\Desktop\ExpenseTracker> python main.py
Enter your choice: 5

----- Monthly Summary -----
2026-06: ₹700.0

===== PERSONAL EXPENSE TRACKER =====
1. Add Expense
2. View All Expenses
3. Category Summary
4. Overall Spending
5. Monthly Summary
6. Exit
Enter your choice: 6
Thank you for using Expense Tracker!
PS C:\Users\denny\OneDrive\Desktop\ExpenseTracker> [ ]
```